

# Bayesian Learning

## Video Lecture 0 - Monte Carlo methods

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# Monte Carlo simulation

- Interest: **expectations of functions**  $f(X)$

$$\mathbb{E}_p(f(X)) = \begin{cases} \sum_{\text{all } x} f(x) \cdot p(x) & \text{for discrete } X \\ \int f(x) \cdot p(x) dx & \text{for continuous } X \end{cases}$$

- Examples:

- ▶ **Mean**  $\mathbb{E}(X)$

$$f(x) = x$$

- ▶ **Variance**  $\mathbb{V}(X)$

$$f(x) = (x - \mu)^2$$

- ▶ **Probabilities**  $\Pr(a \leq X \leq b)$

$$f(x) = \begin{cases} 1 & \text{if } a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

# Properties of sample averages

- **Law of Large Numbers (LLN)**: sample means

$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$  converges to population means  $\mu = \mathbb{E}_p(X)$

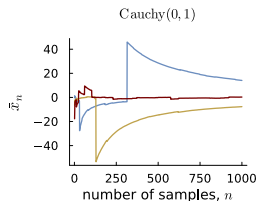
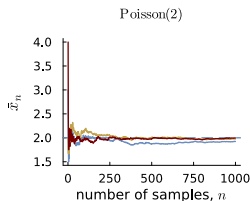
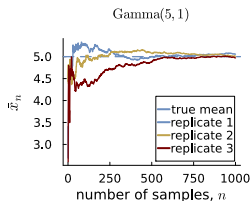
$$\bar{X}_n \xrightarrow{P} \mathbb{E}_p(X)$$

where  $X_1, \dots, X_n$  are iid from  $p(x)$ .

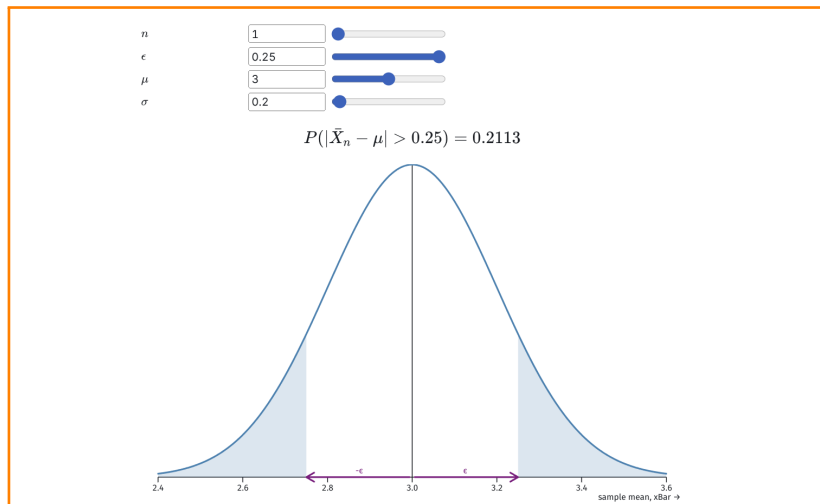
- **Central limit theorem (CLT)**: informally, for large  $n$

$$\bar{X}_n \stackrel{\text{approx}}{\sim} N\left(\mu, \frac{\sigma^2}{n}\right),$$

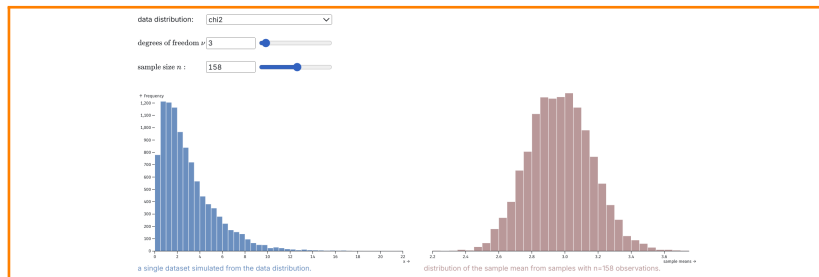
where  $\sigma^2 = \mathbb{V}(X)$ .



# Law of large numbers - widget



# Central limit theorem - widget



# Sample averages of functions

## ■ LLN for a function $f(x)$

$$\bar{f}_n = \frac{1}{n} \sum_{i=1}^n f(X_i) \xrightarrow{p} \mathbb{E}_p(f(X))$$

where  $X_1, \dots, X_n$  are iid from  $p(x)$ .

## ■ CLT for a function $f(x)$

$$\bar{f}_n \stackrel{\text{approx}}{\sim} N\left(\mu_f, \frac{\sigma_f^2}{n}\right),$$

where  $\mu_f = \mathbb{E}_p(f(X))$  and  $\sigma_f^2 = \mathbb{V}_p(f(X))$ .

## ■ We can estimate $\sigma_f^2$ by sample variance of function evaluations

$$\hat{\sigma}_f^2 = \frac{1}{n-1} \sum_{i=1}^n (f(x_i) - \bar{f}_n)^2$$

# Monte Carlo simulation

## ■ Monte Carlo integration

$$\hat{\mu}_f = \frac{1}{m} \sum_{i=1}^m f(x_i), \quad x_i \stackrel{\text{iid}}{\sim} p(x)$$

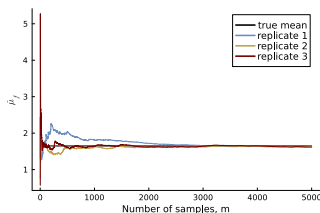
■ **Unbiased**  $\mathbb{E}_p(\hat{\mu}_f) = \mu_f$  for all  $m$ .

■ **LLN: consistent**  $\hat{\mu}_f \xrightarrow{P} \mu_f$  as  $m \rightarrow \infty$ .

■ **CLT**: quantifies **estimation error**  $\hat{\mu}_f - \mu_f$  for finite  $m$ .

■ Example:  $f(x) = \exp(x)$  and  $X \sim N(0, 1)$ .

True mean:  $\mathbb{E}_p(\exp(X)) = \exp(1/2) \approx 1.648$



# Monte Carlo simulation - multivariate

- Two extensions:
  - ▶ Multivariate input:  $\mathbf{x}$
  - ▶ Multivariate output:  $\mathbf{f}(\mathbf{x})$
- Example:  $\mathbf{x} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$  and  $\mathbf{f}(\mathbf{x}) = \mathbf{x}$ .
- **Monte Carlo integration**

$$\hat{\boldsymbol{\mu}}_f = \frac{1}{m} \sum_{i=1}^m \mathbf{f}(\mathbf{x}_i), \quad \mathbf{x}_i \stackrel{\text{iid}}{\sim} p(\mathbf{x})$$

- Still unbiased and consistent.
- Multivariate **CLT**:

$$\hat{\boldsymbol{\mu}}_f \stackrel{\text{approx}}{\sim} N\left(\boldsymbol{\mu}_f, \frac{1}{m} \mathbb{V}_p(\mathbf{f}(X))\right),$$

where  $\mathbb{V}_p(\mathbf{f}(X))$  is a  $d \times d$  covariance matrix.

- Same  $\frac{1}{m}$  rate in multivariate case. Works in high dimensions!



# Importance sampling

- Aim:

$$\mathbb{E}_p(f(X)) = \int f(x) p(x) dx$$

- **Importance sampling (IS)** samples from a **proposal distribution**  $q(x)$  instead of target distribution  $p(x)$ .

- Based on the trivial identity

$$\int f(x) p(x) dx = \int f(x) \frac{p(x)}{q(x)} q(x) dx = \int f(x) w(x) q(x) dx$$

- **Weight function** (importance function)

$$w(x) = \frac{p(x)}{q(x)}$$

- **Importance sampling estimator**

$$\hat{\mu}_f^{\text{IS}} = \frac{1}{m} \sum_{i=1}^m f(x_i) w(x_i), \quad x_i \stackrel{\text{iid}}{\sim} q(x)$$

# Importance sampling

## ■ Importance sampling estimator

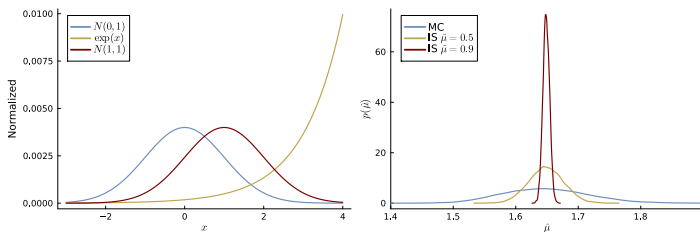
$$\hat{\mu}_f^{\text{IS}} = \frac{1}{m} \sum_{i=1}^m f(x_i) w(x_i), \quad x_i \stackrel{\text{iid}}{\sim} q(x)$$

■ **Unbiased**  $\mathbb{E}_q(\hat{\mu}_f^{\text{IS}}) = \mu_f$  and **consistent**  $\hat{\mu}_f^{\text{IS}} \xrightarrow{P} \mu_f$ .

■ The proposal distribution determines the efficiency (variance).

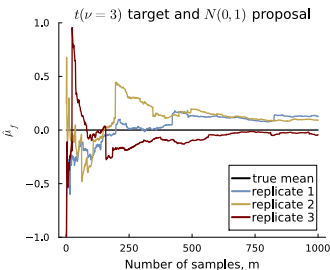
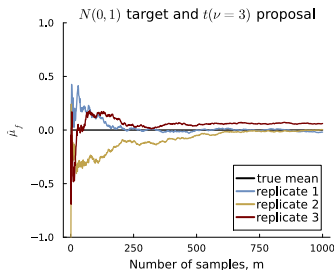
■ Proposal  $q(x)$  proportional to integrand  $f(x) p(x)$  is ideal.

■ Example:  $f(x) = \exp(x)$  and  $X \sim N(0, 1)$ . Proposal:  $N(\tilde{\mu}, 1)$ .



# The tails of the proposal are important

- **Proposal**  $q(x)$  should have **heavier tails** than target  $p(x)$ .
- Example 1: (OK!):
  - ▶ Target:  $N(0, 1)$
  - ▶ Proposal: Student- $t$  with three df.
- Example 2 (not OK!):
  - ▶ Target: Student- $t$  with three df.
  - ▶ Proposal:  $N(0, 1)$



# Rejection sampling

- Rejection sampling samples from the target distribution  $p(x)$ .
- Draws from proposal  $q(x) \neq p(x)$  and then accepts/rejects draws.
- Need to find a majorization constant

$$p(x) \leq M \cdot q(x) \quad \text{for all } x$$

## Rejection sampling:

- 1 Draw a candidate  $x^*$  from the proposal distribution  $q(x)$ .
- 2 Draw  $u \sim \text{Uniform}(0, 1)$ .
- 3 Accept the candidate  $x^*$  if

$$u < \frac{p(x^*)}{M \cdot q(x^*)}$$

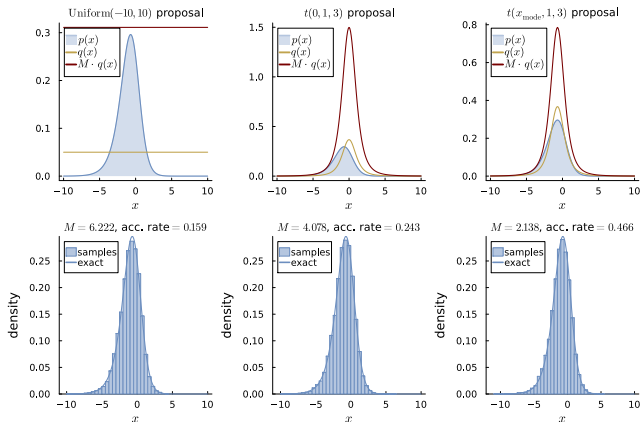
Otherwise, reject the candidate and return to step 1.

# Rejection sampling - example

## ■ Simulation from Logistic-Beta density

$$p(x) = \frac{\sigma(x)^\alpha \sigma(-x)^\beta}{B(\alpha, \beta)}$$

where  $\sigma(x) = 1/(1 + e^{-x})$  is the logistic function.



# Unnormalized target density

- Both Importance sampling and Rejection sampling works with **unnormalized target density**

$$p(x) = \frac{\tilde{p}(x)}{\int \tilde{p}(x) dx}$$

- The unnormalized density  $\tilde{p}(x)$  is known ...
- ... but **normalizing constant**  $\int \tilde{p}(x) dx$  cannot be computed.
- Common in **Bayesian learning** where the target is the **posterior distribution**

$$p(\theta|x) = \frac{p(x|\theta)p(\theta)}{\int p(x|\theta)p(\theta)d\theta}$$

- (Much) more later!